

# MUSTHOFA JOKO ANGGORO

✉ musthofaja.topa@gmail.com |  linkedin.com/in/musthofa-joko-anggoro |  github.com/topahilangharapan |  topahilangharapan.github.io/portfolio

## RESEARCH INTERESTS

Edge AI systems, resource-aware computing in edge-fog-cloud continuum, sustainable computing, energy-efficient systems, distributed systems architectures, hardware-software co-design

## EDUCATION

**University of Indonesia (UI)** Jakarta, Indonesia  
*Bachelor of Computer Science, Information Systems* *Expected Graduation: Aug 2026*  
• **GPA: 3.84/4.00** (8th Semester, Dean's List)

- **Thesis:** Implementation of Morris Mano Computer Architecture with Enhancements on Xilinx Spartan-3 FPGA (X3700AN) using VHDL — demonstrating hardware design, digital logic optimization, and system-level architecture implementation
- **Relevant Coursework:** Computer Architecture, Operating Systems, Data Structures & Algorithms, Introduction to AI & Data Science, Statistics & Probability, Data Communication Networks, Database Systems

**Nanyang Technological University (NTU) & ITS** Singapore & Surabaya  
*INSPIRASI-NTU Summer Program 2025 (LPDP Scholarship Awardee)* *Jan 2025 - Feb 2025*

- Conducted sustainability-focused research on smart port automation and green technology adoption in distributed maritime systems, analyzing IoT sensor networks, automation architectures, and energy-efficient deployments
- Produced comprehensive technical report proposing tiered port automation framework emphasizing renewable energy utilization and carbon emission reduction — directly aligned with sustainability objectives in edge computing systems
- Collaborated with international research teams on applying systems thinking to environmental challenges

## SYSTEMS & RESEARCH EXPERIENCE

**Software Engineer Intern** Traveloka (Travel Technology Company)  
*Systems Engineering & Backend Development* *Aug 2025 - Present*

- Engineered production-grade distributed systems including credential management with AES-GCM encryption, automated rotation snapshots, and granular audit logging using MongoDB for high-throughput operations
- Developed scalable automation services with configurable scrapers and failover mechanisms ensuring 99.9% availability, processing thousands of daily transactions with auto-recovery logic
- Enhanced CI/CD pipelines with automated API testing (Karate framework) and auto-rollback logic to maintain system reliability and prevent production defects in microservice architecture
- Built real-time data synchronization services and optimized structured logging for BigQuery analytics, demonstrating proficiency in distributed data flow and performance optimization
- **Systems Tools:** Java, Spring Boot, MongoDB, Google Cloud Platform, Docker, Kubernetes, BigQuery

**Full Stack Developer** PT Magna Solusi Indonesia  
*Enterprise Systems Development* *Jan 2025 - Jun 2025*

- Developed full-stack enterprise modules with focus on database performance optimization, query efficiency, and backup strategies for high-reliability production systems
- Built RESTful APIs supporting dynamic data flow in distributed web architecture with Spring Boot microservices
- **Systems Tools:** Spring Boot, Angular, PostgreSQL, Apache HTTP Server, Docker, Kubernetes

**Lead Engineer & Full Stack Developer** Event Organizer Enterprise System (Course Project)  
*System Architecture & Implementation* *Feb 2025 - May 2025*

- Led engineering team in designing and implementing enterprise-scale system with focus on clean architecture, secure authentication (JWT), and containerized deployment demonstrating leadership in systems development
- Architected full-stack solution with PostgreSQL optimization, RESTful API design, and CI/CD integration
- **Systems Tools:** Spring Boot, Vue.js, PostgreSQL, Docker, TypeScript

# TECHNICAL SKILLS

---

**Systems Programming:** Java, Python, C (VHDL for hardware), TypeScript, Go  
**Hardware & Architecture:** VHDL, FPGA development (Xilinx Spartan-3), digital logic design, computer architecture implementation  
**Systems & Frameworks:** Spring Boot, Django, microservices architecture, RESTful API design  
**Databases & Storage:** PostgreSQL, MongoDB, manual SQL optimization, query performance tuning  
**DevOps & Infrastructure:** Docker, Kubernetes, CI/CD pipelines, Google Cloud Platform  
**AI/ML Fundamentals:** Scikit-learn, supervised/unsupervised learning, data preprocessing, model evaluation  
**Operating Systems:** Linux, Unix, Windows — familiar with OS internals, process management, memory management

# MACHINE LEARNING PROJECTS

---

|                                  |                                    |
|----------------------------------|------------------------------------|
| <b>Used Car Sales Prediction</b> | Course Project - AI & Data Science |
| <i>Machine Learning Analyst</i>  | <i>Apr 2025 - May 2025</i>         |

- Built comprehensive predictive modeling system for used car sales using imbalanced classification, regression, and clustering techniques to derive actionable business insights from complex datasets
- Implemented multiple ML algorithms (Gradient Boosting, Random Forest, Logistic Regression, Linear Regression, Naive Bayes, K-Means) with emphasis on handling imbalanced data through oversampling and undersampling techniques
- Conducted extensive exploratory data analysis (EDA) and feature engineering, achieving robust model performance through systematic preprocessing, evaluation, and optimization using Scikit-learn
- **Tools & Techniques:** Python, Scikit-learn, Pandas, NumPy, imbalanced learning methods, model evaluation metrics

**Additional Projects:** Database-driven web systems (Django, PostgreSQL with triggers/stored procedures), microservices architecture (Spring Boot), enterprise system development with clean architecture principles